

Optimization 1

ISE 5406

Lecture 3

Dr. Robert Hildebrand

Grado Department of Industrial & Systems Engineering
Virginia Tech

Where We Are in the Course

Last Time (Lecture 2): Introduction to Convexity

- Linear algebra review: combinations, independence, span
- Convex sets: definitions, examples, operations preserving convexity
- Convex and concave functions
- Convex hulls and Carathéodory's theorem
- Extreme points and recession directions

Today (Lecture 3): Real Analysis and Separation

- Closed, open, and compact sets
- Operations preserving closedness and openness
- Existence of minimum/maximum: Weierstrass' theorem
- Separating and supporting hyperplane theorems

Next Time (Lecture 4): Convex Functions

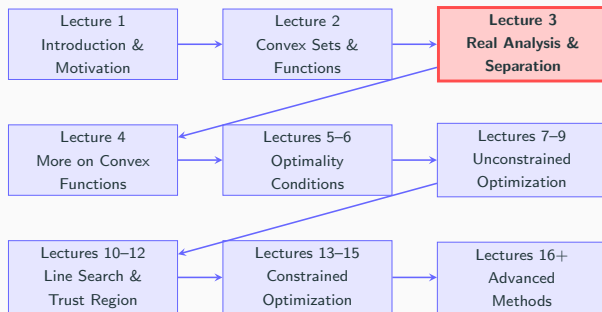
- Strictly convex and strictly concave functions
- Operations preserving convexity of functions
- Continuity of convex functions
- Epigraphs and hypographs

Coming Soon (Lecture 5): Optimality Conditions

- First-order optimality conditions
- Gradients and subgradients
- KKT conditions introduction

Big Picture: Lecture 3 provides the analytical foundations (closed sets, compactness, separation) needed for proving existence of optima and developing optimality conditions.

Big Picture: Course Roadmap



Why Real Analysis and Separation Matter

- **Compactness:** Guarantees existence of optimal solutions (Weierstrass)
- **Separation theorems:** Foundation for duality and KKT conditions
- **Closedness:** Essential for proving convergence of algorithms

Reading for this lecture

- Chapter 2 of Bazaraa, Sherali, and Shetty (2006)
- Chapter 2 and Appendix A of Boyd and Vandenberghe

Outline

Real Analysis

Closed, Open, and Compact Sets

Operations Preserving Closedness and Openness of Sets

Existence of Minimum and Maximum

Infimum and Supremum

Weierstrass' Theorem

Separating Hyperplanes

Closest-Point Theorem

Separating Hyperplane Theorem

Supporting Hyperplanes

Supporting Hyperplane Theorem

Roadmap

Real Analysis

Closed, Open, and Compact Sets

Operations Preserving Closedness and Openness of Sets

Existence of Minimum and Maximum

Infimum and Supremum

Weierstrass' Theorem

Separating Hyperplanes

Closest-Point Theorem

Separating Hyperplane Theorem

Supporting Hyperplanes

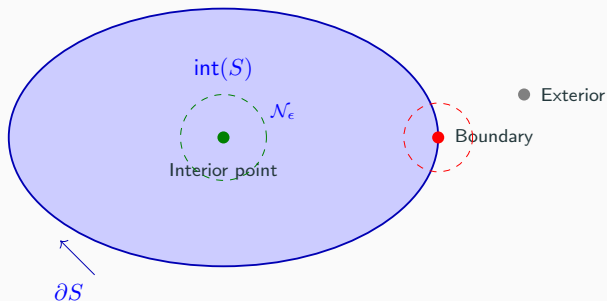
Supporting Hyperplane Theorem

Definitions: Real Analysis

Let S be an arbitrary set in $\mathbb{R}^n$. (Most definitions extend beyond $\mathbb{R}^n$.)

- **Closure:** $\mathbf{x} \in \mathbb{R}^n$ is said to be in the closure of S , denoted $\text{cl}(S)$, if $S \cap \mathcal{N}_\epsilon(\mathbf{x}) \neq \emptyset$ for every $\epsilon > 0$.
- **Interior:** $\mathbf{x} \in \mathbb{R}^n$ is said to be in the interior of S , denoted $\text{int}(S)$, if $\mathcal{N}_\epsilon(\mathbf{x}) \subset S$ for some $\epsilon > 0$. A solid set S is one that has $\text{int}(S) \neq \emptyset$.
- **Closed Set:** If $S = \text{cl}(S)$, then S is called closed.
- **Open Set:** If $S = \text{int}(S)$, then S is called open.
- **Bounded Set:** S is said to be bounded if it can be contained in a ball of a sufficiently large radius centered at the origin.
- **Compact Set:** S is compact if it is both closed and bounded.
- **Boundary:** $\mathbf{x} \in \mathbb{R}^n$ is said to be on the boundary of S , denoted ∂S , if for every $\epsilon > 0$, $\mathcal{N}_\epsilon(\mathbf{x})$ contains at least one point in S and at least one point not in S .

Visual Summary: Closure, Interior, and Boundary



- Interior $\text{int}(S)$: All points with a neighborhood entirely inside S
- Boundary ∂S : Points where every neighborhood touches both S and S^c
- Closure $\text{cl}(S) = \text{int}(S) \cup \partial S$

Sequences in $\mathbb{R}^n$

Definition (Sequence)

A **sequence** in $\mathbb{R}^n$ is a function from $\mathbb{N}$ (or $\mathbb{Z}_+$) to $\mathbb{R}^n$. We denote a sequence as $\{\mathbf{x}_k\}_{k=1}^{\infty}$ or simply $\{\mathbf{x}_k\}$, where $\mathbf{x}_k \in \mathbb{R}^n$ for each k .

Definition (Convergent Sequence)

A sequence $\{\mathbf{x}_k\}$ in $\mathbb{R}^n$ **converges** to a point $\bar{\mathbf{x}} \in \mathbb{R}^n$ if for every $\epsilon > 0$, there exists an integer K such that

$$\|\mathbf{x}_k - \bar{\mathbf{x}}\| < \epsilon \quad \text{for all } k \geq K.$$

We write $\lim_{k \rightarrow \infty} \mathbf{x}_k = \bar{\mathbf{x}}$ or $\mathbf{x}_k \rightarrow \bar{\mathbf{x}}$.

Intuition: The sequence converges to $\bar{\mathbf{x}}$ if the terms eventually get (and stay) arbitrarily close to $\bar{\mathbf{x}}$.

Subsequences

Definition (Subsequence)

A **subsequence** of $\{\mathbf{x}_k\}$ is a sequence $\{\mathbf{x}_{k_j}\}_{j=1}^{\infty}$ where $k_1 < k_2 < k_3 < \dots$ is a strictly increasing sequence of positive integers.

Example: If $\{\mathbf{x}_k\} = \{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5, \dots\}$, then

- $\{\mathbf{x}_{2j}\} = \{\mathbf{x}_2, \mathbf{x}_4, \mathbf{x}_6, \dots\}$ (even-indexed terms) is a subsequence
- $\{\mathbf{x}_{k^2}\} = \{\mathbf{x}_1, \mathbf{x}_4, \mathbf{x}_9, \mathbf{x}_{16}, \dots\}$ is a subsequence

Definition (Convergent Subsequence)

A subsequence $\{\mathbf{x}_{k_j}\}$ **converges** to $\bar{\mathbf{x}}$ if for every $\epsilon > 0$, there exists J such that $\|\mathbf{x}_{k_j} - \bar{\mathbf{x}}\| < \epsilon$ for all $j \geq J$.

Key Fact: If $\mathbf{x}_k \rightarrow \bar{\mathbf{x}}$, then every subsequence also converges to $\bar{\mathbf{x}}$.

But: A sequence may have convergent subsequences even if the sequence itself diverges!

Bolzano–Weierstrass Theorem

Theorem (Bolzano–Weierstrass)

Every **bounded** sequence in $\mathbb{R}^n$ has a **convergent subsequence**.

Example: $\{(-1)^k\} = \{-1, 1, -1, 1, \dots\}$ is bounded but does not converge.

- Subsequence of odd terms: $\{-1, -1, -1, \dots\} \rightarrow -1$
- Subsequence of even terms: $\{1, 1, 1, \dots\} \rightarrow 1$

Why This Matters:

- Provides a key tool for proving existence results in optimization
- Used in proving Weierstrass' theorem (existence of min/max on compact sets)
- Bounded sequences may not converge, but always have “limit points”

Connection to Compactness

A set S is compact if and only if every sequence in S has a subsequence that converges to a point in S . (Sequential characterization of compactness in $\mathbb{R}^n$.)

Proof: Bolzano–Weierstrass Theorem

Theorem (Bolzano–Weierstrass)

Every **bounded** sequence in $\mathbb{R}^n$ has a **convergent subsequence**.

Proof:

Lemma (Closed Sets)

Suppose $S \subseteq \mathbb{R}^n$ is a closed set. Then

1. S contains all its boundary points, i.e., $\partial S \subseteq S$.
2. Its complement $\mathbb{R}^n \setminus S$ is open.
3. For any convergent sequence $\{\mathbf{x}_k\}$ in S with $\lim_k \mathbf{x}_k = \bar{\mathbf{x}}$, we have $\bar{\mathbf{x}} \in S$.

Lemma (Open Sets)

A set $S \subseteq \mathbb{R}^n$ is open if and only if it does not contain any of its boundary points, i.e., $\partial S \cap S = \emptyset$.

Lemma (Boundary)

The boundary of a set $S \subseteq \mathbb{R}^n$ is given by $\partial S = \text{cl}(S) \setminus \text{int}(S)$.

Proof: Lemma (Closed Sets)

Lemma (Closed Sets)

Suppose $S \subseteq \mathbb{R}^n$ is a closed set. Then

1. S contains all its boundary points, i.e., $\partial S \subseteq S$.
2. Its complement $\mathbb{R}^n \setminus S$ is open.
3. For any convergent sequence $\{\mathbf{x}_k\}$ in S with $\lim_k \mathbf{x}_k = \bar{\mathbf{x}}$, we have $\bar{\mathbf{x}} \in S$.

Proof:

Proof: Lemma (Open Sets)

Lemma (Open Sets)

A set $S \subseteq \mathbb{R}^n$ is open if and only if it does not contain any of its boundary points, i.e., $\partial S \cap S = \emptyset$.

Proof:

Proof: Lemma (Boundary)

Lemma (Boundary)

The boundary of a set $S \subseteq \mathbb{R}^n$ is given by $\partial S = \text{cl}(S) \setminus \text{int}(S)$.

Proof:

Exercise: Classifying Sets

Classification Challenge

For each set below, determine: (1) Is it open? (2) Is it closed? (3) What is its interior? (4) Is it compact?

Set	Open/Closed/Compact?
$\mathbb{R}^n$ and $\emptyset$	
$\{\mathbf{x} \in \mathbb{R}^n : \sum_{i=1}^n x_i^2 \leq 1\}$ (closed ball)	
$\{\mathbf{x} \in \mathbb{R}^n : \sum_{i=1}^n x_i^2 < 1\}$ (open ball)	
$\{\mathbf{x} \in \mathbb{R}^n : \sum_{i=1}^n x_i^2 = 1\}$ (sphere)	
Hyperplane $\{\mathbf{x} : \mathbf{a}^T \mathbf{x} = b\}$	
Halfspace $\{\mathbf{x} : \mathbf{a}^T \mathbf{x} \leq b\}$	

Solution: Classifying Sets

Answers

Set	Classification
$\mathbb{R}^n$ and $\emptyset$	Both open AND closed (only sets with this property!)
Closed ball $\{\mathbf{x} : \ \mathbf{x}\ ^2 \leq 1\}$	Closed, not open. Interior = open ball. Compact.
Open ball $\{\mathbf{x} : \ \mathbf{x}\ ^2 < 1\}$	Open, not closed. Not compact (not closed).
Sphere $\{\mathbf{x} : \ \mathbf{x}\ ^2 = 1\}$	Closed, not open. $\text{int}(S) = \emptyset$. Compact.
Hyperplane $\{\mathbf{x} : \mathbf{a}^T \mathbf{x} = b\}$	Closed, not open. $\text{int}(S) = \emptyset$. Not compact (unbounded).
Halfspace $\{\mathbf{x} : \mathbf{a}^T \mathbf{x} \leq b\}$	Closed, not open. Not compact (unbounded).

Operations Preserving Closedness and Openness of Sets

Lemma

1. The intersection of any (even infinite) number of closed sets is closed.
2. The intersection of *finitely* many open sets is open.
3. The union of *finitely* many closed sets is closed.
4. The union of any (even infinite) number of open sets is open.

Can you think of *counterexamples* to the following assertions?

1. The intersection of infinitely many open sets is open.
2. The union of infinitely many closed sets is closed.

Supplemental Reference: Walter Rudin's "Principles of Mathematical Analysis"

Exercise: Counterexamples

Think-Pair-Share

Find counterexamples to show that:

1. The intersection of infinitely many **open** sets may not be open.
2. The union of infinitely many **closed** sets may not be closed.

Hint: Consider sets of the form $(-1/n, 1/n)$ or $[1/n, 1 - 1/n]$ for $n = 1, 2, 3, \dots$

Solution: Counterexamples

Answers

1. Intersection of open sets may not be open:

Consider $S_n = (-1/n, 1/n)$ for $n = 1, 2, 3, \dots$ (all open intervals).

$$\bigcap_{n=1}^{\infty} S_n = \{0\}$$

The intersection is a single point, which is **closed, not open**.

2. Union of closed sets may not be closed:

Consider $S_n = [1/n, 1 - 1/n]$ for $n = 2, 3, 4, \dots$ (all closed intervals).

$$\bigcup_{n=2}^{\infty} S_n = (0, 1)$$

The union is an open interval, which is **open, not closed**.

Convexity and Closure/Interior

Theorem

Let S be a convex set in $\mathbb{R}^n$ with a nonempty interior. Let $\mathbf{x}_1 \in \text{cl}(S)$ and $\mathbf{x}_2 \in \text{int}(S)$. Then

$$\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 \in \text{int}(S) \text{ for each } \lambda \in (0, 1).$$

Proof: reading assignment (see Theorem 2.2.2 of Bazaraa et al.)

Corollaries

1. Let S be a convex set. Then $\text{int}(S)$ is convex.
2. Let S be a convex set with $\text{int}(S) \neq \emptyset$. Then $\text{cl}(S)$ is convex.
3. Let S be a convex set with $\text{int}(S) \neq \emptyset$. Then $\text{cl}(\text{int}(S)) = \text{cl}(S)$.
4. Let S be a convex set with $\text{int}(S) \neq \emptyset$. Then $\text{int}(\text{cl}(S)) = \text{int}(S)$.

We can replace $\text{int}(S)$ with the relative interior of S in the above results.

Roadmap

Real Analysis

Closed, Open, and Compact Sets

Operations Preserving Closedness and Openness of Sets

Existence of Minimum and Maximum

Infimum and Supremum

Weierstrass' Theorem

Separating Hyperplanes

Closest-Point Theorem

Separating Hyperplane Theorem

Supporting Hyperplanes

Supporting Hyperplane Theorem

Infimum and Supremum

Let S be an arbitrary set in $\mathbb{R}$.

Infimum: A scalar $l \in \mathbb{R}$ is called the infimum or the **greatest lower bound** of S , denoted $\inf(S)$, if

1. l is a lower bound of S , i.e., $l \leq x$ for each $x \in S$,
2. for all lower bounds b of S , $l \geq b$.

Convention: $\inf(\emptyset) = +\infty$, $\inf(S) = -\infty$ if S is unbounded from below

Supremum: A scalar $u \in \mathbb{R}$ is called the supremum or the **least upper bound** of S , denoted $\sup(S)$, if

1. u is an upper bound of S , i.e., $u \geq x$ for each $x \in S$,
2. for all upper bounds b of S , $u \leq b$.

Convention: $\sup(\emptyset) = -\infty$, $\sup(S) = +\infty$ if S is unbounded from above

Exercise: Infimum vs Minimum

Quick Check

For each set, find the infimum and determine if the minimum exists.

1. $S = (0, 1]$

2. $S = [0, 1]$

3. $S = \{1/n : n \in \mathbb{N}\}$

4. $S = \{x \in \mathbb{R} : x^2 > 2\}$

Solution: Infimum vs Minimum

Answers

1. $S = (0, 1]$: $\inf(S) = 0$, but minimum does not exist (0 is not in S).
2. $S = [0, 1]$: $\inf(S) = 0$, and $\min(S) = 0$ exists (0 is in S).
3. $S = \{1/n : n \in \mathbb{N}\} = \{1, 1/2, 1/3, \dots\}$:
 $\inf(S) = 0$, but minimum does not exist (0 is not achieved by any $1/n$).
4. $S = \{x \in \mathbb{R} : x^2 > 2\} = (-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$:
 $\inf(S) = -\infty$ (unbounded below), so minimum does not exist.

Key insight: The infimum always exists (in the extended reals), but the minimum exists only if the infimum is attained by some element of S .

Existence of Minimum and Maximum

Let $S \subseteq \mathbb{R}^n$ be a nonempty set and $f : S \rightarrow \mathbb{R}$.

When do the problems

$$\inf_{\mathbf{x} \in S} f(\mathbf{x}) / \min_{\mathbf{x} \in S} f(\mathbf{x}) \quad \text{and} \quad \sup_{\mathbf{x} \in S} f(\mathbf{x}) / \max_{\mathbf{x} \in S} f(\mathbf{x})$$

admit (optimal) solutions?

- $\bar{\mathbf{x}}$ is a minimizing solution for the problem $\inf\{f(\mathbf{x}) : \mathbf{x} \in S\}$, provided that $\bar{\mathbf{x}} \in S$ and $f(\bar{\mathbf{x}}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in S$.

If a minimizing solution exists, we simply write $\min_{\mathbf{x} \in S} f(\mathbf{x})$.

- $\bar{\mathbf{x}}$ is a maximizing solution for the problem $\sup\{f(\mathbf{x}) : \mathbf{x} \in S\}$, provided that $\bar{\mathbf{x}} \in S$ and $f(\bar{\mathbf{x}}) \geq f(\mathbf{x})$ for all $\mathbf{x} \in S$.

If a maximizing solution exists, we simply write $\max_{\mathbf{x} \in S} f(\mathbf{x})$.

Continuity of Functions

Definition (Continuous Function)

A function $f : S \rightarrow \mathbb{R}$ is **continuous at** $\bar{x} \in S$ if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$\|x - \bar{x}\| < \delta \text{ and } x \in S \quad \Rightarrow \quad |f(x) - f(\bar{x})| < \epsilon.$$

We say f is **continuous on** S if it is continuous at every point in S .

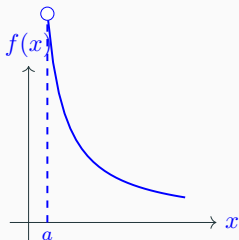
Sequential Characterization of Continuity

$f : S \rightarrow \mathbb{R}$ is continuous at $\bar{x} \in S$ if and only if for every sequence $\{x_k\}$ in S with $x_k \rightarrow \bar{x}$, we have $f(x_k) \rightarrow f(\bar{x})$.

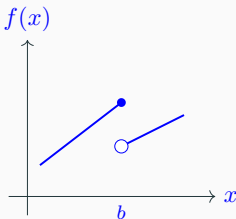
Intuition:

- Continuity means “small changes in input produce small changes in output”
- The sequential characterization says “ f preserves limits”
- This characterization is key for proving Weierstrass’ theorem

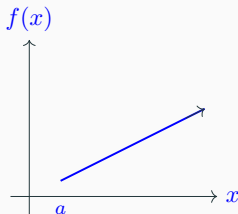
Why Optima May Not Exist



(a) S not closed



(b) f discontinuous



(c) S unbounded

(a) S is not closed: infimum not attained at boundary

(b) f is not continuous: jump at discontinuity

(c) S is unbounded: function keeps decreasing (or increasing)

Weierstrass' Theorem

Weierstrass' Theorem (Sufficient Conditions for Existence of Min./Max.)

Let S be a nonempty, **compact** (closed and bounded) set, and let

$f : S \rightarrow \mathbb{R}$ be **continuous** on S . Then the problem $\min\{f(\mathbf{x}) : \mathbf{x} \in S\}$ attains its minimum.

Key Points:

- Compactness = closed + bounded
- Continuity of f is essential
- The theorem guarantees *existence*, not uniqueness
- Same result holds for maximum

Why This Matters for Optimization:

- Provides conditions under which we *know* an optimal solution exists
- Justifies writing “**min**” instead of “**inf**” when conditions are met
- Foundation for convergence analysis of optimization algorithms

Proof: Weierstrass' Theorem

Weierstrass' Theorem

Let S be a nonempty, compact set, and let $f : S \rightarrow \mathbb{R}$ be continuous on S . Then the problem $\min\{f(\mathbf{x}) : \mathbf{x} \in S\}$ attains its minimum.

Proof:

Exercise: Applying Weierstrass' Theorem

Application

For each optimization problem, determine if Weierstrass' theorem guarantees a solution exists. If not, explain which condition fails.

1. $\min\{x^2 : x \in [-1, 1]\}$

2. $\min\{1/x : x \in (0, 1]\}$

3. $\min\{e^{-x} : x \in [0, \infty)\}$

4. $\min\{x_1^2 + x_2^2 : x_1^2 + x_2^2 \leq 4\}$

Solution: Applying Weierstrass' Theorem

Answers

1. $\min\{x^2 : x \in [-1, 1]\}$

Yes, solution exists. $S = [-1, 1]$ is compact, $f(x) = x^2$ is continuous. $\min = 0$ at $x = 0$.

2. $\min\{1/x : x \in (0, 1]\}$

No guarantee. $S = (0, 1]$ is **not closed** (missing 0). Indeed, $\inf = 1$ but no minimum.

3. $\min\{e^{-x} : x \in [0, \infty)\}$

No guarantee. $S = [0, \infty)$ is **not bounded**. Indeed, $\inf = 0$ but no minimum (approaches 0 as $x \rightarrow \infty$).

4. $\min\{x_1^2 + x_2^2 : x_1^2 + x_2^2 \leq 4\}$

Yes, solution exists. S is the closed ball of radius 2 (compact), f is continuous. $\min = 0$ at origin.

Roadmap

Real Analysis

Closed, Open, and Compact Sets

Operations Preserving Closedness and Openness of Sets

Existence of Minimum and Maximum

Infimum and Supremum

Weierstrass' Theorem

Separating Hyperplanes

Closest-Point Theorem

Separating Hyperplane Theorem

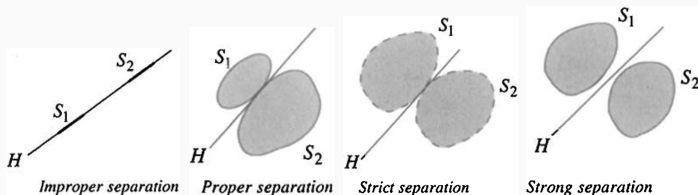
Supporting Hyperplanes

Supporting Hyperplane Theorem

Separating Hyperplane

Let S_1 and S_2 be nonempty sets in $\mathbb{R}^n$.

- A hyperplane $H := \{\mathbf{x} \in \mathbb{R}^n : \mathbf{p}^T \mathbf{x} = \alpha\}$ **separates** S_1 and S_2 if $\mathbf{p}^T \mathbf{x} \geq \alpha$ for each $\mathbf{x} \in S_1$ and $\mathbf{p}^T \mathbf{x} \leq \alpha$ for each $\mathbf{x} \in S_2$.
- If, additionally, $S_1 \cup S_2 \not\subset H$, then H **properly separates** S_1 and S_2 .
- The hyperplane H **strictly separates** S_1 and S_2 if $\mathbf{p}^T \mathbf{x} > \alpha$ for each $\mathbf{x} \in S_1$ and $\mathbf{p}^T \mathbf{x} < \alpha$ for each $\mathbf{x} \in S_2$.
- The hyperplane H **strongly separates** S_1 and S_2 if for some $\epsilon > 0$, $\mathbf{p}^T \mathbf{x} \geq \alpha + \epsilon$ for each $\mathbf{x} \in S_1$ and $\mathbf{p}^T \mathbf{x} \leq \alpha$ for each $\mathbf{x} \in S_2$.



Exercise: Types of Separation

Conceptual Question

Consider the following pairs of sets in $\mathbb{R}^2$. For each pair, determine the strongest type of separation possible (none, improper, proper, strict, or strong).

1. $S_1 = \{(x, y) : x \geq 0\}$ and $S_2 = \{(x, y) : x \leq 0\}$

2. $S_1 = \{(x, y) : x > 0\}$ and $S_2 = \{(x, y) : x < 0\}$

3. $S_1 = \{(x, y) : x \geq 1\}$ and $S_2 = \{(x, y) : x \leq 0\}$

Solution: Types of Separation

Answers

1. $S_1 = \{(x, y) : x \geq 0\}$ and $S_2 = \{(x, y) : x \leq 0\}$

The hyperplane $x = 0$ separates them, but $S_1 \cup S_2 = \mathbb{R}^2 \supset H$.

Improper separation (both sets share the boundary).

2. $S_1 = \{(x, y) : x > 0\}$ and $S_2 = \{(x, y) : x < 0\}$

The hyperplane $x = 0$ separates them with $\mathbf{p}^T \mathbf{x} > 0$ for S_1 and < 0 for S_2 .

Strict separation (but not strong—no gap ϵ).

3. $S_1 = \{(x, y) : x \geq 1\}$ and $S_2 = \{(x, y) : x \leq 0\}$

Take $x = 0.5$. Then $\mathbf{p}^T \mathbf{x} \geq 1 \geq 0.5 + 0.5$ for S_1 and $\leq 0 < 0.5$ for S_2 .

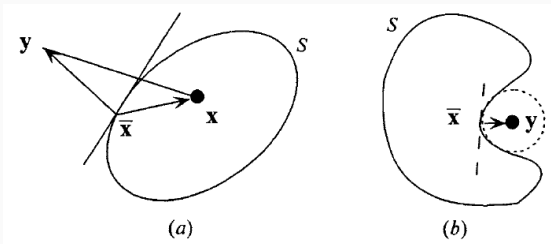
Strong separation (there's a gap of width 1 between the sets).

Closest-Point Theorem

Closest-Point Theorem

Given a nonempty closed convex set $S \subseteq \mathbb{R}^n$ and a point $\mathbf{y} \in \mathbb{R}^n$ with $\mathbf{y} \notin S$, there exists a unique point $\bar{\mathbf{x}} \in S$ with minimum distance from $\mathbf{y}$. Moreover, $\bar{\mathbf{x}}$ is the minimizing point if and only if

$$(\mathbf{y} - \bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}}) \leq 0, \text{ for all } \mathbf{x} \in S.$$



Exercise: Closest Point

Computation

Find the closest point in S to the given point $\mathbf{y}$.

1. $S = \{\mathbf{x} \in \mathbb{R}^2 : x_1 + x_2 \leq 1, x_1 \geq 0, x_2 \geq 0\}, \mathbf{y} = (2, 2)^T$

2. $S = \{\mathbf{x} \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1\}, \mathbf{y} = (3, 4)^T$

Hint: For problem 2, the closest point lies on the boundary in the direction of $\mathbf{y}$ from the origin.

Solution: Closest Point

Answers

1. $S = \{\mathbf{x} \in \mathbb{R}^2 : x_1 + x_2 \leq 1, x_1 \geq 0, x_2 \geq 0\}, \mathbf{y} = (2, 2)^T$

The closest point is on the hypotenuse $x_1 + x_2 = 1$.

Project $(2, 2)$ onto the line $x_1 + x_2 = 1$:

Direction: $\mathbf{n} = (1, 1)/\sqrt{2}$. Distance to line:

$$(2 + 2 - 1)/\sqrt{2} = 3/\sqrt{2}.$$

$$\bar{\mathbf{x}} = (2, 2) - \frac{3}{\sqrt{2}} \cdot \frac{(1, 1)}{\sqrt{2}} = (2, 2) - (1.5, 1.5) = (0.5, 0.5)^T$$

2. $S = \{\mathbf{x} \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1\}, \mathbf{y} = (3, 4)^T$

The closest point is on the boundary (unit circle) in the direction of $\mathbf{y}$.

$$\|\mathbf{y}\| = \sqrt{9 + 16} = 5$$

$$\bar{\mathbf{x}} = \frac{\mathbf{y}}{\|\mathbf{y}\|} = (3/5, 4/5)^T = (0.6, 0.8)^T$$

Separating Hyperplane Theorem

Theorem (Separation of a Convex Set and a Point)

Let S be a nonempty closed convex set in $\mathbb{R}^n$ and $\mathbf{y} \notin S$. Then, there exists a nonzero vector $\mathbf{p}$ and a scalar α such that

$$\mathbf{p}^T \mathbf{y} > \alpha \text{ and } \mathbf{p}^T \mathbf{x} \leq \alpha \text{ for each } \mathbf{x} \in S.$$

Corollaries

1. Let S be a closed convex set in $\mathbb{R}^n$. Then S is the intersection of all halfspaces containing S .
2. Let S be a nonempty set, and let $\mathbf{y} \notin \text{cl}(\text{conv}(S))$. Then there exists a strongly separating hyperplane separating S and $\{\mathbf{y}\}$.
3. **Farkas' Lemma:** Let $A \in \mathbb{R}^{m \times n}$ and $\mathbf{c} \in \mathbb{R}^n$. Then exactly one of the following two systems has a solution:

System 1: $A\mathbf{x} \leq \mathbf{0}$ and $\mathbf{c}^T \mathbf{x} > 0$ for some $\mathbf{x} \in \mathbb{R}^n$.

System 2: $A^T \mathbf{y} = \mathbf{c}$ and $\mathbf{y} \geq \mathbf{0}$ for some $\mathbf{y} \in \mathbb{R}^m$.

Proof: Separating Hyperplane Theorem

Theorem (Separation of a Convex Set and a Point)

Let S be a nonempty closed convex set in $\mathbb{R}^n$ and $\mathbf{y} \notin S$. Then, there exists a nonzero vector $\mathbf{p}$ and a scalar α such that

$$\mathbf{p}^T \mathbf{y} > \alpha \text{ and } \mathbf{p}^T \mathbf{x} \leq \alpha \text{ for each } \mathbf{x} \in S.$$

Proof:

Roadmap

Real Analysis

Closed, Open, and Compact Sets

Operations Preserving Closedness and Openness of Sets

Existence of Minimum and Maximum

Infimum and Supremum

Weierstrass' Theorem

Separating Hyperplanes

Closest-Point Theorem

Separating Hyperplane Theorem

Supporting Hyperplanes

Supporting Hyperplane Theorem

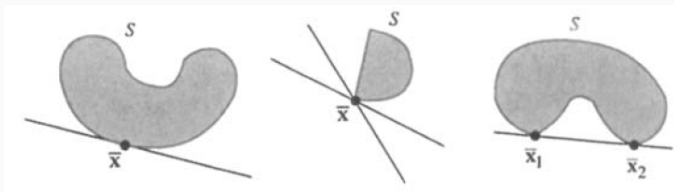
Supporting Hyperplane

Let S be a nonempty set in $\mathbb{R}^n$ and $\bar{\mathbf{x}} \in \partial S$.

- A hyperplane $H := \{\mathbf{x} \in \mathbb{R}^n : \mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) = 0\}$ is called a **supporting hyperplane** of S at $\bar{\mathbf{x}}$ if S is *entirely* contained in one of the halfspaces

$$\{\mathbf{x} \in \mathbb{R}^n : \mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0\}, \quad \text{or} \quad \{\mathbf{x} \in \mathbb{R}^n : \mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0\}$$

- If $S \not\subseteq H$, then H is a **proper supporting hyperplane** of S at $\bar{\mathbf{x}}$.



Supporting Hyperplane Theorem

Theorem (Supporting Hyperplane)

Let S be a nonempty convex set in $\mathbb{R}^n$ and $\bar{\mathbf{x}} \in \partial S$. Then there exists a hyperplane that supports S at $\bar{\mathbf{x}}$. That is, there exists $\mathbf{p} \neq \mathbf{0}$ such that $\mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$ for each $\mathbf{x} \in \text{cl}(S)$.

Geometric Interpretation:

- At every boundary point of a convex set, we can find a hyperplane that “touches” the set without crossing into it
- The set lies entirely on one side of the supporting hyperplane
- This is a key property that characterizes convex sets

Application: Supporting hyperplanes are fundamental to:

- Characterizing optimal solutions (KKT conditions)
- Duality theory
- Cutting plane methods in optimization

Exercise: Supporting Hyperplanes

Geometric Construction

For each set and boundary point, find a supporting hyperplane.

1. $S = \{\mathbf{x} \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1\}$ at $\bar{\mathbf{x}} = (1, 0)^T$

2. $S = \{\mathbf{x} \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq 0, x_1 + x_2 \leq 1\}$ at $\bar{\mathbf{x}} = (0, 0)^T$

3. $S = \{\mathbf{x} \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq 0, x_1 + x_2 \leq 1\}$ at $\bar{\mathbf{x}} = (0.5, 0.5)^T$

Solution: Supporting Hyperplanes

Answers

1. $S = \{\mathbf{x} : x_1^2 + x_2^2 \leq 1\}$ at $\bar{\mathbf{x}} = (1, 0)^T$

The gradient of $g(\mathbf{x}) = x_1^2 + x_2^2 - 1$ at $(1, 0)$ is $(2, 0)^T$.

Supporting hyperplane: $x_1 = 1$ (or $\mathbf{p} = (1, 0)^T$).

2. $S = \{\mathbf{x} : x_1 \geq 0, x_2 \geq 0, x_1 + x_2 \leq 1\}$ at $\bar{\mathbf{x}} = (0, 0)^T$

At the corner $(0, 0)$, multiple supporting hyperplanes exist!

Any $\mathbf{p}$ with $p_1 \leq 0$ and $p_2 \leq 0$ works.

Examples: $x_1 = 0$, $x_2 = 0$, or $x_1 + x_2 = 0$.

3. $S = \{\mathbf{x} : x_1 \geq 0, x_2 \geq 0, x_1 + x_2 \leq 1\}$ at $\bar{\mathbf{x}} = (0.5, 0.5)^T$

This point is on the edge $x_1 + x_2 = 1$.

Supporting hyperplane: $x_1 + x_2 = 1$ (or $\mathbf{p} = (1, 1)^T$).

Quick Assessment (5 questions)

Answer True or False:

1. **T/F:** Every bounded set is compact.
2. **T/F:** The interior of a convex set is convex.
3. **T/F:** Weierstrass' theorem guarantees that $\min\{1/x : x \in (0, 1]\}$ attains its minimum.
4. **T/F:** Every pair of disjoint convex sets can be strictly separated.
5. **T/F:** At every boundary point of a convex set, there exists a supporting hyperplane.

Answers

1. **False.** Compact = closed AND bounded. Example: $(0, 1)$ is bounded but not closed, hence not compact.
2. **True.** This is Corollary 1 from the theorem about convex sets and interior/closure.
3. **False.** $(0, 1]$ is not closed (hence not compact), so Weierstrass doesn't apply. Indeed, $\inf = 1$ but the minimum is not attained.
4. **False.** Consider $S_1 = \{(x, y) : x < 0\}$ and $S_2 = \{(x, y) : x > 0\} \cup \{(0, 0)\}$. They're disjoint but can only be improperly separated.
5. **True.** This is exactly the Supporting Hyperplane Theorem.

Summary and Next Steps

Key Takeaways

- **Closed sets** contain all boundary points; **open sets** contain none
- **Compact** = closed + bounded (essential for existence of optima)
- **Weierstrass' theorem**: Continuous function on compact set attains min/max
- **Separating hyperplanes**: Can separate a point from a closed convex set
- **Supporting hyperplanes**: Exist at every boundary point of a convex set

Next Time:

- Strictly convex and strictly concave functions
- Operations preserving convexity of functions
- Epigraphs and their connection to convex sets

Questions?